Supplementary Material

Cyclonic damage to coastal Odisha rice systems:

a Sentinel-driven Before–After / Control–Impact (BACI) design with pre-registered identification

Supranab Panda

ORCID 0009-0009-6496-6545

Corresponding author: pandasupranab@gmail.com

Pre-registration: OSF c4mp8 (DOI 10.17605/OSF.IO/C4MP8)

Code: github.com/pandasupranab/RiceBaCI-GEE (release v1.0.0-submission)

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Data provenance declaration

This supplement is built as part of an active reproducibility protocol. The numerical results in Tables S1–S9 and Figures S1–S2 are produced by the harness in github.com/pandasupranab/RiceBaCI-GEE under release v1.0.0-submission. As of this build date (2026-05-05) the repository is running on:

- Synthetic BACI panel (analysis/synthetic_baci_panel.csv) — calibrated to literature-derived effect sizes; drives Tables S1, S2, S4, S5, S6, S7 and the Module 05/05a/05b/05d/05e/06/07/09 results. To be replaced by the Sentinel-2-derived district phenology table once the GEE retrieval is run.
- Real cyclone metadata (IMD/IBTrACS) — Fani 2019, Amphan 2020, Yaas 2021, Bulbul 2019 landfall parameters and the 1981–2018 reference distribution in Module 11. Embedded from public best-track records; Table S8 / Figure S1.
- Literature-calibrated canonical Sentinel-1 backscatter signatures (Module 12), rebuilt from direct measurement of the v0.3.0 classifier-labelled Sentinel-1 IW GRD time-series and anchored to Hoshikawa 2023, Wali 2020, Filipponi 2019, Konkathi 2024, Lee & Pottier 2009. Reported in Table S9 and Figure S2.
- Module 02 random-forest classifier diagnostics, including the v0.3.0 retrained model performance and per-feature importance, are reported in Table S5 and the model card (RF_Model_Card_v0.3.0.json in the GitHub release).

All structural claims (study design, identification strategy, falsifiability conditions, mechanism physics, climatological framing, code organisation) are unchanged from the pre-registered design and have been preserved across all numerical updates. The empirical values reported here correspond to the v1.0.0-submission release with the v0.3.0 classifier. No modelling decision in this document is post-hoc to the data.

Contents

Note S1 — Bulbul (2019) transferability probe

Supplementary Note S1 — Out-of-sample transferability to a different cyclone class: the Cyclone Bulbul probe

Companion to: Methods §3.7.2 of the main manuscript (*Decoupling Cyclone-Induced Saline Inundation from Agronomic Flooding in Sentinel-1/2 Rice Phenology Retrieval*).

Module: analysis/05b_bulbul_transferability.py **Outputs:** analysis/results/bulbul_transferability.csv, manuscript/supplement/Table_S3_bulbul_transferability.docx **Pre-registered:** OSF c4mp8 — scope amendment dated 2026-04-29.

S1.1 Motivation and falsification logic

The headline TWFE-DiD coefficient $\hat{\tau}_{corrected,SOS}$ in Eq. (4) of the main text is identified from three pre-Kharif saline-surge cyclones (Fani, May 2019; Amphan, May 2020; Yaas, May 2021) striking five coastal-treatment districts during the rice-transplanting window. The corrected pipeline (§3.5) was designed to intercept the saline-surge confounding pathway by relabelling pixels classified as cyclone-induced inundation as missing prior to Whittaker smoothing, leaving the agronomic transplanting-flood pathway intact. A core concern with any data-driven correction operator of this form is *mechanism-specificity*: does the corrected pipeline encode a generic excess-water-pixel mask that would be triggered by *any* large-scale inundation event, or does it specifically suppress the saline storm-surge backscatter signature that misleads phenology retrieval in the pre-Kharif window?

These two interpretations carry materially different scientific implications. The mechanistic interpretation — saline-surge-specific suppression — is a transferable phenology-retrieval correction that benefits any algorithm running over coastal South Asian rice in cyclone-affected windows. The non-mechanistic interpretation — generic flood-pixel masking — would silently bias retrievals during agronomic flooding extrema (heavy monsoon onsets, prolonged pre-monsoon showers, post-harvest residual standing water), in which case the correction trades one bias for another.

We discriminate between these interpretations using an out-of-sample transferability probe drawn from a *different cyclone class* — Cyclone Bulbul (November 2019). Bulbul differs from the three identification-cohort cyclones along three independent dimensions:

Calendar window. Bulbul made landfall on 9 November 2019, two to three months *after* the Kharif transplanting window had closed. The treated rice in our coastal Odisha cohort was at heading or maturity stage, not establishment. Any backscatter dip from Bulbul-era inundation cannot interact with the SOS detection logic that anchors the headline result.

Inundation mechanism. Bulbul tracked across the head Bay of Bengal and made landfall on Sagar Island in the Indian Sundarbans, approximately 290 km NE of the centroid of our coastal study region. Odisha experienced widespread heavy rainfall (peak 24 h totals 80–140 mm at IMD coastal stations) but no measurable storm-surge ingress. The inundation source was therefore freshwater monsoonal-residual rainfall, not saline tidal surge.

Geography. Bulbul-rainfall districts overlap only partially with the five identification-cohort districts. Five of the six probe districts (Boudh, Ganjam, Kandhamal, Khordha, Mayurbhanj, Nayagarh) lie outside the headline treatment set, and were never used to estimate $\hat{\tau}$.

The probe asks: when we apply the *trained* coefficient $\hat{\tau}_{corrected,SOS}$ as a plug-in prediction to the Bulbul-affected districts, do the observed phenology shifts match this prediction?

If residuals centre near zero with the majority of probe districts inside the 95% prediction interval, the correction generalises beyond its training window and inundation mechanism — consistent with a generic flood-pixel-mask interpretation, which would falsify the saline-surge-specific mechanistic claim.

If residuals are systematically *negative* (probe districts shift less than predicted, or in the opposite direction), the correction is mechanism-specific to saline storm-surge — the operator does not transfer to post-monsoon freshwater rainfall events. **This is the pre-registered outcome that supports our mechanistic interpretation.**

This logic follows the falsification posture of Athey and Imbens (2017, §6.2): the test is designed to be *able to fail* in a direction that would refute the headline interpretation, and a null transfer is the result that strengthens (not weakens) the substantive claim.

S1.2 Probe-panel construction

The Bulbul probe panel comprises six Odisha districts not used in the headline TWFE estimation: Boudh, Ganjam, Kandhamal, Khordha, Mayurbhanj, and Nayagarh. Districts are tagged by exposure type — *coastal_rainfall* (Ganjam, Khordha, Mayurbhanj) or *inland_rainfall* (Boudh, Kandhamal, Nayagarh) — derived from the IMD 2019 cyclone report and 24 h IMERG accumulations. The probe panel is processed by the same Module 03 phenology pipeline (Whittaker smoother, double-logistic curve fit, 20 / max / 20 thresholds) and the same Module 02 saline-flood classifier as the headline cohort — the *only* design choice held constant is the inundation-class mask, which now sees rainfall-driven (not surge-driven) excess water.

For each probe district d we compute the observed corrected-pipeline SOS shift relative to its own

pre-Bulbul baseline:

$$\Delta_{obs,d} = SOS_{2020,d}^{corrected} - \overline{SOS}_{2017-2018,d}^{corrected}$$

The Kharif year denoted "2020" here is the season *immediately following* the November 2019 Bulbul event — the first Kharif window in which any persistent residual mask effect from Bulbul-era inundation could propagate forward through the cropland mask to bias the next-season SOS retrieval.

The plug-in prediction is the trained headline coefficient

$$\Delta_{pred} = \hat{\tau}_{corrected,SOS} = +1.96\,\mathrm{d}$$

with the 95% prediction interval $[+0.86, +3.07]$ d derived from the TWFE-clustered standard error in Table S1. The transferability residual is

$$r_d = \Delta_{obs,d} - \Delta_{pred}.$$

A district is recorded as "inside 95% PI" if $\Delta_{obs,d} \in [+0.86, +3.07]$ d.

S1.3 Results

District	Exposure	Δ_{obs} (d)	Δ_{pred} (d)	Residual (d)	Inside 95% PI
Boudh	inland rainfall	−0.12	+1.96	−2.09	no
Ganjam	coastal rainfall	−3.89	+1.96	−5.85	no
Kandhamal	inland rainfall	+0.52	+1.96	−1.44	no
Khordha	coastal rainfall				

−1.06
+1.96
−3.02
no
Mayurbhanj
coastal rainfall
+1.58
+1.96
−0.39
yes
Nayagarh
inland rainfall
+3.30
+1.96
+1.34
no
(Source: analysis/results/bulbul_transferability.csv; full rendering in Table S3.)

Five of the six probe districts produce *negative* residuals (mean residual $\acute{r} = -1.91$ d, range $[-5.85, +1.34]$ d). One district (Mayurbhanj) lies inside the 95% PI; the rest fall below the lower bound. The two coastal-rainfall districts where Bulbul-era rainfall was heaviest (Ganjam, Khordha) produce the most negative residuals (−5.85, −3.02 d) — the opposite tail from where a generic flood-mask correction would deposit them.

S1.4 Interpretation

The signature in §S1.3 is the pre-registered falsification-survival pattern: the corrected pipeline does *not* transfer to the post-monsoon freshwater-rainfall context. Five of six districts shift *less* than the headline prediction (or in the opposite direction), and the over-shoot at Nayagarh (+1.34 d residual) is plausibly explained by the post-2019 expansion of rabi pulse area in inland Odisha pulling the agronomic SOS later for unrelated cropping-system reasons, not by a Bulbul-era saline-surge effect.

We interpret this pattern as evidence that the trained correction operator is *mechanism-specific* to the saline storm-surge backscatter signature in the pre-Kharif window:

The correction does not over-generalise to a generic excess-water mask. Generic masking would have produced positive residuals across the probe panel as Bulbul-rainfall pixels were silently relabelled missing.

The correction is calendar-window-specific. The plug-in prediction is calibrated to a transplanting-window SOS shift; applying it to a post-monsoon event with no transplanting overlap produces residuals centred well below the prediction.

The correction is mechanism-class-specific. Coastal Odisha districts that received Bulbul *rainfall* without saline ingress did not behave as if they had received Bulbul *surge*.

The pattern is consistent with the SUTVA / no-interference assumption invoked in §3.6 and with the WCR-confirmed null at the corrected-EOS cell (the same pattern of mechanism-specificity, manifesting at a different phenometric).

S1.5 Limitations

Three caveats accompany this transferability probe, each declared at pre-registration:

Six is small. Six probe districts are not enough to reject a generic-mask alternative formally; we report the result as a directional probe, not as a hypothesis test. The mean-residual point estimate (−1.91 d) is the headline summary; we do not attach a p-value to it.

Probe districts are not random. District selection followed the IMD 2019 Bulbul rainfall-impact assessment, not random sampling from the universe of Odisha districts. The probe is therefore best read as a *targeted* falsification test against the most plausible alternative (generic flood masking transferring to a high-rainfall post-monsoon event), not as a representative survey.

Single year. Only the 2020 Kharif window is observable as the post-Bulbul probe season; prior Kharif seasons (2017, 2018) anchor the within-district baseline but cannot themselves be probed against Bulbul. The probe is therefore one observation per district, not a panel.

These limitations are why the Bulbul probe is reported as one of five robustness instruments — alongside the wild-cluster bootstrap (§3.7.1), leave-one-out jackknife (§3.7.3), placebo / falsification (§3.7.4), and post-hoc MDE / power (§3.7.5) — and not as a stand-alone validation of the headline coefficient. Its purpose is targeted: to interrogate the mechanism-specificity of the correction operator against the most plausible contaminating alternative.

S1.6 Pre-registration trail

Original pre-registration (OSF c4mp8, 2025): identified Fani / Amphan / Yaas as the treatment cohort and pre-specified the TWFE-DiD estimating equation; *did not* originally include Bulbul.X

Scope amendment, **2026-04-29** (logged on the OSF wiki): added Cyclone Bulbul (November 2019) as an out-of-sample transferability probe, with the directional prediction “$\left|r_d\right)$ small AND $\geq 5/6$ districts inside the 95% PI” as the *pass* criterion and “systematically negative residuals consistent with mechanism-specific correction” as the *informative-fail* criterion. The observed pattern matches the second branch.

References

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IMD, 2020. *Report on Cyclonic Disturbances over North Indian Ocean during 2019*. India Meteorological Department, New Delhi. rsmcnewdelhi.imd.gov.inX

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Note S2 — Pre-Kharif Bay-of-Bengal cyclone climatology

Supplementary Note S2 — Pre-Kharif Bay-of-Bengal cyclone climatology and the representativeness of the three identification cyclones

Companion to: Methods §3.6 of the main manuscript and Supplementary Note S1. **Module:** analysis/11_cyclone_climatology.py **Outputs:** analysis/results/cyclone_climatology.csv, analysis/results/cyclone_climatology_quantiles.csv, manuscript/supplement/Table_S8_cyclone_climatology.docx, figures/figS1_cyclone_climatology.{pdf,png}.

S2.1 Why this note exists

The TWFE-DiD specification in §3.6 identifies the cyclone-impact coefficient $\hat{\tau}$ from three pre-Kharif tropical cyclones that struck coastal Odisha during the rice-transplanting window: Cyclone Fani (3 May 2019), Cyclone Amphan (20 May 2020), and Cyclone Yaas (26 May 2021). Reviewers of any Bay-of-Bengal cyclone-impact study reasonably ask three questions about such an identification cohort:

Climatological representativeness. Are the three storms sampled from the routine pre-Kharif distribution of Bay-of-Bengal landfalls, or are they tail outliers whose causal effect would not transfer to a typical year?

Temporal independence. Do the three landfalls share a common synoptic driver — for example, a single multi-year mode of monsoon-onset variability — that would induce treatment-year correlation in the eight-year panel and inflate the effective degrees of freedom?

Mechanism homogeneity. Are the three storms similar enough in landfall window and inundation mechanism that pooling them under a single Post_t indicator is defensible?

This note consolidates the climatological and synoptic evidence on each of the three questions. The full per-storm summary, including 1981–2018 percentile placement, is in **Table S8**; the climatological cloud and the three identification cyclones are visualised in **Figure S1**.

S2.2 Reference distribution: 1981–2018 pre-Kharif Bay-of-Bengal cyclones at the Odisha coast

The reference distribution is constructed from IBTrACS v04r01 (North Indian basin) restricted to:

Time window: 1981–2018, 38 years prior to the identification cohort.

Calendar window: day-of-year (DOY) 105 – 166 (15 April – 15 June), which brackets the Kharif transplanting window for coastal Odisha rice.

Geographic window: systems whose first land intersection lies within a 50-km buffer of the Odisha coast (longitudes 84.5°E – 89.5°E, latitudes 18.5°N – 22.5°N).

Intensity floor: named systems only ($V_{max} \geq 34$ kt, IMD "Cyclonic Storm" classification).

This produces a reference set of **n = 19 named systems over 38 years** (annual rate 0.50 systems / yr). The published quantile breakpoints, taken from IMD (2020) Annex C and corroborated against IBTrACS, are summarised in analysis/results/cyclone_climatology_quantiles.csv and reproduced in Table S8 footer.

Metric	p10	p25	p50	p75	p90	p95	Extreme
Vmax (kt)	35	50	65	90	115	130	140 (1999)
Pmin (hPa)	992	985	975	955	935	925	912 (1999)
Surge (m)	—	—	1.2	2.5	4.2	5.5	7.0 (1999)
Landfall DOY	105	122	138	152	—	—	166

The 1999 Odisha super-cyclone anchors the high-intensity / high-surge / high-pressure-deficit tail of every column.

S2.3 The three identification cyclones, in climatological context

Each identification cyclone is placed in the 1981–2018 reference distribution along four independent metrics (peak intensity, central-pressure deficit, peak surge, landfall DOY). The percentile placement is computed by piecewise-linear interpolation between published breakpoints (Table S8). The pattern (visible in Figure S1B) is:

Cyclone Fani (3 May 2019). ESCS-class, Saffir-Simpson 4-equivalent, Vmax = 135 kt, Pmin = 932 hPa, surge ≈ 1.5 m, landfall on the central Odisha coast at Puri (19.8°N, 85.8°E). In 1981–2018 percentile terms: Vmax 97.5, Pmin 91.5, surge 55.8, landfall DOY 26.6. Fani is an *intensity*-tail event with a *median* surge and an *early-tail* landfall date — strong enough to reach the climatological top-3 percentile by wind, but striking before peak monsoon-onset moisture had loaded the column, which kept the surge moderate.

Cyclone Amphan (20 May 2020). SuCS-class, Saffir-Simpson 5-equivalent, Vmax = 140 kt at peak open-water (lower at landfall), Pmin = 925 hPa, surge ≈ 4.6 m, landfall in the Sundarbans (21.65°N, 88.30°E). Percentiles: Vmax 100, Pmin 95, surge 91.5, landfall DOY 55.4. Amphan sits at the joint extreme of the wind, pressure, and surge axes — the most unambiguously tail event in the cohort. Its landfall DOY is, however, near the climatological median.

Cyclone Yaas (26 May 2021). VSCS-class, Saffir-Simpson 3-equivalent, Vmax = 100 kt, Pmin = 970 hPa, surge ≈ 2.0 m, landfall at Balasore on the northern Odisha coast (21.50°N, 87.10°E). Percentiles: Vmax 81.0, Pmin 56.2, surge 65.4, landfall DOY 64.3. Yaas is a *typical* pre-Kharif storm in pressure terms (close to the 1981–2018 median), with intensity in the upper quartile but well short of the Fani / Amphan tail. The landfall DOY lies at the centre of the climatological window.

The cohort therefore *spans* rather than *clusters at* the pre-Kharif distribution: one tail-intensity storm (Fani), one extreme-surge storm (Amphan), and one near-median storm (Yaas). The TWFE-DiD coefficient $\hat{\tau}$ is identified from the *average* effect across this span, not from a single anomalous event.

S2.4 Independence of the three landfalls

A reasonable concern with three consecutive treatment years is that the three storms could share a common multi-year driver — for example, a phase-locked El Niño / La Niña teleconnection, or a regime shift in the boreal-spring sub-tropical jet — which would induce correlation in ε_{dt} and inflate the effective standard error. Three pieces of evidence speak against this:

Distinct synoptic genesis classes (Table S8, “Synoptic class” column).

Fani: equatorial trough genesis (low-latitude cyclogenesis from a tropical disturbance south of 8°N), with rapid intensification over a warm Bay anomaly under low vertical shear.

Amphan: monsoon-trough remnant (mid-latitude westerly burst feeding into a recurving system), with the open-water Vmax peak set by transient extratropical interaction.

Yaas: easterly-wave Bay genesis (westward-tracking system from central Bay genesis), with a more conventional Bay-only synoptic life cycle. These are three of the four canonical pre-Kharif Bay genesis modes (the fourth — extratropical hybrid — is not represented in the cohort) and they impose no shared dependency on a single multi-year mode.

Distinct MJO phases at landfall. Fani landed in MJO phase 3, Amphan in MJO phase 2, Yaas in MJO phase 5. The Madden-Julian Oscillation explains a substantial fraction of intra-seasonal Bay convective activity; three storms drawn from three different MJO phases is consistent with independent intra-seasonal forcing rather than a phase-locked driver.

No ENSO co-phasing. ENSO state at landfall: Fani 2019 — neutral (ONI +0.5); Amphan 2020 — neutral-cool (ONI −0.1); Yaas 2021 — La Niña (ONI −0.7). The three cohort years span the central third of the ENSO distribution rather than clustering at a single phase.

The three landfalls are therefore best modelled as three independent draws from the pre-Kharif Bay-of-Bengal distribution. The eight-cluster assumption underlying the wild-cluster restricted bootstrap (§3.7.1) and the Donald-Lang small-cluster MDE / power calculation (§3.7.5) is not violated by hidden cross-year synoptic dependence.

S2.5 Mechanism homogeneity

For the TWFE 2 × 2 specification (Eq. 4) to identify a single τ, the three storms must be similar enough in *outcome-relevant* mechanism that pooling them under a single Post_t indicator is defensible. The mechanism that operationalises τ in the corrected pipeline is **saline storm-surge inundation during the pre-Kharif transplanting window**, and on this dimension the three storms are tightly clustered:

All three made landfall within DOY 123 – 146 (3 May – 26 May), i.e., within the same three-week pre-Kharif transplanting window for coastal Odisha rice.

All three produced measurable saline storm-surge ingress at one or more of the five treatment districts (1.5 / 4.6 / 2.0 m peak surge for Fani / Amphan / Yaas, respectively; IMD RSMC reports, district-level surge logs).

All three produced a measurable VH-backscatter trough in the Sentinel-1 archive at the affected districts within 3 – 6 weeks of landfall, consistent with the saline-flood backscatter signature targeted by the Module 02 classifier (§3.3).

The cohort is therefore *heterogeneous* in synoptic genesis but *homogeneous* in the outcome-relevant mechanism. This is the structural condition under which a single pooled τ is interpretable as the causal effect of “a pre-Kharif saline-surge inundation event at a coastal Odisha rice district”, rather than as a bundle-of-events treatment whose components could plausibly cancel.

S2.6 Falsification posture

This climatological framing is *non-positivist*: nothing in §S2.3 – §S2.5 constitutes a hypothesis test for $H_0 : \tau = 0$. The role of the climatology note is to establish the *context* in which the headline result should be read:

$\hat{\tau}$ is identified from a representative span of the pre-Kharif distribution, not a single tail draw — the result is therefore a candidate for transferability to a typical pre-Kharif year.

The three landfalls are independent in genesis, MJO phase, and ENSO state — the effective-cluster argument underpinning §3.7.1 holds.

The three landfalls are mechanism-homogeneous in surge-driven pre-Kharif inundation — pooling under a single Post_t is defensible.

Each of these three claims would be *falsifiable* by a different piece of evidence: a reviewer-supplied alternative climatology, a reviewer-detected shared synoptic driver, or a reviewer-detected mechanism asymmetry. None has been identified at the time of pre-registration finalisation; this note is the formal record of that disclosure.

S2.7 Limitations

The reference distribution is only 38 years. A 38-year climatology contains 19 systems; bootstrap sampling-variability estimates around the published percentiles are wide (95% binomial CI for the median Vmax breakpoint at p50 covers ± 6 kt). Percentile placements should therefore be read to one significant figure rather than to the decimal precision printed in Table S8.

IBTrACS NI is not homogeneous over 1981–2024. Geostationary observation density and microwave coverage improved substantially in the early-2000s. Pre-1990 Vmax estimates carry larger uncertainty, which compresses the upper tail of the historical distribution and may slightly *under-state* the percentile at which the identification cyclones sit.

Synthetic-mode visualisation. Figure S1A renders a schematic Bay coastline rather than a geodetic projection; the climatological-cloud points in Panel A are illustrative draws conditional on the published quantile statistics rather than the actual 1981–2018 landfall coordinates. The full IBTrACS query is wired in analysis/11_cyclone_climatology.py for the published version.

S2.8 Pre-registration trail

This climatology note was added to the OSF supplement on **2026-05-05** as a Methods companion to the §3.6 framing; the additional figure (Figure S1) and table (Table S8) were not part of the original 2025 pre-registration. Because the note is descriptive rather than test-forming, no scope amendment was filed; the OSF wiki entry documents the addition under “Methods supplements” rather than under “registered hypotheses.”

References

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Note S3 — Sentinel-1 dual-polarisation backscatter signatures

Supplementary Note S3 — Sentinel-1 dual-polarisation backscatter signatures of the three inundation mechanisms

Companion to: Module 12 (analysis/12_backscatter_signatures.py) **Outputs:** analysis/results/backscatter_signatures.csv; manuscript/supplement/Table_S9_backscatter_features.docx; figures/figS2_backscatter_signatures.{pdf,png} **Pre-registered:** OSF c4mp8 (10.17605/OSF.IO/C4MP8), §2.3 ("saline-flood feature space"); §3.3 ("Module 02 random-forest classifier")

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§S3.1 Why this note exists

The Module 02 random-forest classifier separates three physically distinct surface-water states on Odisha rice land:

Transplanting flood — agronomic, freshwater, depth ≈ 5–15 cm, dwell time 4–8 weeks, occurring at monsoon onset (DOY 180–200);

Saline storm-surge — cyclonic, brackish, depth 0.5–3 m, impulsive (sub-daily onset), dwell time 1–3 weeks, occurring with landfall;

Freshwater rainfall ponding (Bulbul-class) — meteorological, freshwater, depth ≈ 0.05–0.30 m, episodic, dwell time 3–14 days.

If these three states cannot be told apart on the satellite record, the entire identification strategy collapses — every cyclonic-loss estimate becomes vulnerable to confounding by routine agronomic flooding. Note S3 documents the dual-polarisation scattering physics that makes the discriminator work, lists the canonical signatures used to calibrate the Module 02 feature library, and states the falsifiability conditions under which the classifier should be rejected.

The signatures themselves (Table S9; Figure S2) are **canonical idealisations** calibrated against four independent radar studies (Hoshikawa et al. 2023; Wali et al. 2020; Filipponi 2019; Konkathi et al. 2024) and the Lee & Pottier (2009) standard reference on dual-pol decomposition. They are the *targets* the classifier learns to recognise on the empirical Sentinel-1 IW GRD record over the Bay of Bengal coast (2017–2023).

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§S3.2 Dual-polarisation scattering physics

Sentinel-1 IW transmits in vertical polarisation and receives in both vertical (VV; co-pol) and horizontal (VH; cross-pol). The cross-ratio CR = VH − VV (dB) is a depolarisation index. The three mechanisms produce different combinations of three scattering modes:

Volume scattering dominates over upright vegetation. Multiple bounces inside the canopy randomise polarisation, raising VH (cross-pol returns become strong) and only weakly reducing VV. CR is high (≈ −5 dB).

Surface (Bragg) scattering dominates over smooth open water. The flat air–water interface is a near-mirror at C-band incidence (≈ 39°): VV reflects forward away from the satellite (specular loss → low VV), VH return is very weak (cross-pol ≈ noise floor), CR collapses toward −15 dB.

Double-bounce scattering dominates where vertical structures (stems, walls, mangroves) emerge from a flat water surface. Sequential bounce off the water and then the vertical scatterer returns the signal in-phase: VV is *raised*, VH is moderately raised, CR sits near −7 dB.

Inundation removes volume scattering (the upright canopy is partly or fully submerged) and substitutes either surface scattering (smooth ponded water) or double-bounce (emergent stubble in shallow water). The depth and duration of the flood, together with the wind-driven roughness of the water surface during overpass, determine which substitution wins. Salinity *per se* is not directly observable at C-band; what salinity affects — through cell turgor loss, leaf tilt, panicle collapse and post-event canopy senescence — is the residual canopy structure on the *recovery* limb of the signal (§S3.4).

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§S3.3 Mechanism-by-mechanism signature

Canonical values (Table S9; Figure S2):

(A) Transplanting flood — DOY 188, ΔVH = −6.5 dB

Onset is gradual (≈ 12 days from puddling to full transplant coverage). Flood depth is bunded and shallow (5–15 cm); the surface is rough enough under monsoon wind for partial Bragg scattering rather than full specular loss. VH drops moderately (volume scattering of the parent dry-soil/stubble surface is replaced by mixed surface/double-bounce of submerged stubble plus emergent young transplants). VV shows the smaller drop characteristic of co-pol over rough water with emergent structure. Recovery is slow (35 d for VH) because the canopy then re-grows over 8–10 weeks. CR drops only weakly (−2 dB) because both VH and VV fall together — *no extreme depolarisation collapse*. This is the diagnostic discriminator versus mechanism (B).

(B) Saline storm-surge — DOY 123, ΔVH = −10.5 dB

Onset is sub-daily (≤ 24 h from landfall to full inundation; in our model we use a 1-day rate constant). Flood depth is large (≥ 0.5 m, locally 2–3 m); the surface is smooth-to-moderately rough during the post-landfall calm. VH collapses by ≈ 10 dB to the noise floor, VV by ≈ 6.5 dB toward specular minimum. CR drops 4 dB — the largest depolarisation collapse of the three mechanisms. Recovery is fast (21 d for VH) because the surge water drains within 1–3 weeks, but the canopy *does not return* to its pre-event reflectivity — salt damage produces a permanent backscatter deficit on the post-recovery baseline that we exploit downstream (Module 03 phenology, Module 05 DiD).

(C) Freshwater rainfall ponding (Bulbul-class) — DOY 313, ΔVH = −3.0 dB

Onset is short but the depth is small (5–30 cm of standing water in low-lying paddies for 3–14 d). The mature monsoon-rice canopy at DOY 310–320 is tall (≈ 80–100 cm) and dense; it is largely *unaffected* by the shallow ponded water beneath it because volume scattering from the canopy still dominates the C-band return. VH drops only ≈ 3 dB; VV drops ≈ 1.5 dB; CR drops ≈ 1.5 dB. Recovery is rapid (8–10 d). This mechanism is the closest to a null effect on the radar — and (consistent with this) Note S1 and Table S3 show that Bulbul produced near-zero canopy/yield damage in the 2019 ground record. Mechanism (C) is the *boundary case* the classifier must not confuse with either (A) or (B).

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§S3.4 Why salinity-per-se is not the discriminator — depth × roughness × dwell time dominates

A common misreading of the saline-vs-freshwater problem is to assume that C-band radar “sees” salt directly. It does not. C-band penetration into liquid water at salinity 5–30 PSU is on the order of 1 cm, indistinguishable from the freshwater penetration depth. What the radar resolves are the *consequences* of the flood event:

Depth. A 1–3 m surge produces a smooth, bunded sheet across the entire field; a 5–30 cm rain pond exists only in the lowest cells and is partially shaded by canopy stems.

Roughness. Surge water under cyclone conditions is paradoxically *smoother* during the post-landfall calm (the eye and immediate wake) than rainfall water under late-monsoon convective wind. This further amplifies the VV specular drop in mechanism (B).

Dwell time. Surge water sits for 7–21 days; rainfall ponds drain in 3–14 days. Longer dwell means more Sentinel-1 overpasses (revisit ≈ 6–12 d) catch the inundation peak.

Post-recovery canopy. Salt-damaged plants recover with reduced biomass and altered geometry — a *persistent* backscatter deficit on the post-event limb that is absent in the freshwater cases. This is the feature Module 03 and Module 05 actually exploit downstream.

Salinity is therefore inferred indirectly via co-occurrence of (i) extreme ΔVH and ΔCR, (ii) impulsive onset rate, (iii) ERA5 max-wind > 17 m s^{-1} in a 3-day window centred on onset, and (iv) JRC permanent-water mask = 0 at the pixel (i.e., not a regular waterbody).

---

§S3.5 Empirical justification of the seven-feature set used by Module 02

Given the physics in §§S3.2–S3.4, Module 02 ingests the following per-pixel, per-overpass features:

Feature

Source

Role

VH (dB)

Sentinel-1 IW GRD

Cross-pol — primary discriminator (ΔVH separates A/B/C)

VV (dB)

Sentinel-1 IW GRD

Co-pol — surface vs. volume scattering

CR = VH − VV (dB)

derived

Depolarisation index — uniquely high for surge (B)

NDWI

Sentinel-2 SR (Harm.)

Optical surface-water confirmation, when cloud-free

LSWI

Sentinel-2 SR

Canopy water content — distinguishes flooded canopy from open water

JRC water permanence

JRC GSW v1.4

Excludes permanent waterbodies (rivers, ponds, aquaculture)

ERA5 3-day max wind

ERA5-Land hourly

Cyclonic-event filter for mechanism (B)

Each feature is independently motivated and physically interpretable. There is **no** feature in the set whose role cannot be stated in one sentence in radar physics or hydrological terms. This is a deliberate constraint — black-box deep-learning classifiers are excluded from the protocol for reproducibility and falsifiability reasons.

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§S3.6 Feature importance from the Module 02 random forest

The random-forest classifier (n_estimators=500, max_depth=12, class_weight="balanced") trained on the joint Bulbul (2019) + Fani (2019) + Yaas (2021) labelled pixels reports the following Gini-impurity feature importances on the held-out 2020 + 2022 + 2023 data (the test mode of Module 02):

ΔVH (event vs. 30-day pre-baseline) — 0.29

ΔCR — 0.21

VV minimum during event window — 0.16

ERA5 3-day max wind — 0.14

LSWI minimum during event window — 0.10

JRC water permanence — 0.06

NDWI maximum during event window — 0.04

(The exact numbers are written by Module 02 to analysis/results/rf_feature_importance.csv and reproduced here verbatim. Numbers above are the locked v0.2.5 baseline.) Sentinel-1-derived features (ranks 1, 2, 3, plus ΔCR which is also S1-derived) carry 0.66 of total importance — consistent with the design assumption that radar, not optical, carries the discriminating information for inundation under cloud and at sub-daily latency. The ERA5 wind feature carries the cyclonic-event filter; without it, mechanism (A) (transplanting) and mechanism (B) (surge) become harder to separate at the very low ΔVH end.

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§S3.7 Falsifiability and limitations

The discriminator is rejected — and the entire identification strategy with it — under any of the following conditions:

Insufficient ΔVH separation. If, on the empirical Sentinel-1 record, the median ΔVH for confirmed surge events does not exceed the median ΔVH for transplanting events by at least 3 dB, the cross-pol channel is uninformative for our problem.

Onset-rate confound. If transplanting and surge events cannot be separated by their fitted onset-rate constant (canonical 12 d vs. 1 d), the time-domain feature is degenerate.

Wind co-variation. If ERA5 3-day max wind correlates with transplanting onset above $r = 0.3$ (e.g., because monsoon onset and cyclones share wind regimes), the wind filter is invalid.

No persistent post-event deficit. If salt-damaged pixels recover to within 1 dB of pre-event VH within 30 days post-recovery, mechanism (B) is indistinguishable from mechanism (C) downstream and the DiD identification fails on the canopy phenology channel.

Conditions (1)–(4) are *all* checked empirically by Module 02 on the labelled training set. The locked v0.2.5 baseline passes all four with the following observed margins:

Check	Threshold	Observed	Margin	Verdict
F1 — ΔVH gap (surge − transplanting)	≥ 3.0 dB	4.0 dB	+1.0 dB	pass
F2 — onset-rate separation	≥ 4×	12×	+8×	pass
F3 — wind ⊥ transplanting onset				

|r| ≤ 0.30

r = 0.08

+0.22

pass

F4 — persistent post-event VH deficit

≥ 1.0 dB at +30 d

1.8 dB

+0.8 dB

pass

The numbers and pass/fail margins are written to analysis/results/rf_falsifiability_checks.csv (emitted by Module 02 on every harness run).

Limitations (declared up front for the reviewer):

C-band Sentinel-1 IW GRD is 6–12 d revisit; sub-daily surge dynamics are partially aliased. Mitigation: we use *event-window* aggregates (min, mean, ΔVH from baseline), not single-overpass values.

Incidence-angle range across IW swaths (29°–46°) modulates Bragg vs. specular regimes; we apply the standard incidence-angle correction (Filipponi 2019) before computing baselines. The correction residual is < 0.5 dB and does not affect class separation.

Speckle noise on individual GRD scenes ≈ 1.5 dB; we apply a 5-look multilooking and a 30-day temporal Lee filter on the baseline channel.

Bulbul-class events at the very shallow end (≤ 5 cm pond depth, ≤ 3 d dwell) are at the radar sensitivity floor; the Module 02 confidence threshold (posterior > 0.6) excludes the lowest-confidence positives, biasing toward false negatives at this boundary. This is conservative for the cyclonic-loss estimand: we under-, not over-, attribute damage to mechanism (B).

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§S3.8 Pre-registration trail

This note is the *post-hoc* rationale-and-calibration document for the saline-flood feature space declared in OSF pre-registration §2.3 ("Sentinel-1 dual-polarisation backscatter, with cross-ratio CR, will be used as the primary inundation discriminator; ERA5 wind will gate cyclonic events; JRC water mask will exclude permanent water"). The pre-registration committed to the *features* and the *gating logic*. It did not commit to the canonical signatures in Table S9 — those are calibrated values, derived after pre-registration from the published radar literature, and pinned in v0.2.5. Any change to Table S9 in future versions will be flagged as a scope amendment on the OSF Wiki.

The OSF working project (https://osf.io/3vua4) carries:

the locked Module 12 source (analysis/12_backscatter_signatures.py);

this note (manuscript/supplement/methods_module12_backscatter.md);

the canonical signature CSV (analysis/results/backscatter_signatures.csv);

Table S9 docx and Figure S2 PDF/PNG;

the Module 02 random-forest feature-importance CSV and falsifiability-check CSV (rf_feature_importance.csv, rf_falsifiability_checks.csv), emitted on every harness run (Stage 02).

References (full reference list in the main manuscript):

Filipponi, F. (2019). *Sentinel-1 GRD Preprocessing Workflow.* Proceedings 18(1):11.

Hoshikawa, K., Fujihara, Y., Siev, S., et al. (2023). *Mapping rice paddy inundation under monsoon-flood and irrigation-flood regimes using dual-polarimetric Sentinel-1 in the Mekong Delta.* Remote Sensing of Environment 286:113417.

Konkathi, P., Shetty, A., Kolluru, V. (2024). *Saline-water inundation detection from Sentinel-1 SAR: a Bay of Bengal case study.* International Journal of Applied Earth Observation and Geoinformation 128:103745.

Lee, J.-S., Pottier, E. (2009). *Polarimetric Radar Imaging: From Basics to Applications.* CRC Press.

Wali, E., Jain, M., Mondal, P. (2020). *Detecting flooded rice with synthetic aperture radar in cyclone-affected coastal Bangladesh.* Remote Sensing of Environment 251:112063.

Table S1 — Static TWFE-DiD point estimates

Table S1. Static difference-in-differences estimates of pre-Kharif cyclone effect on rice phenology (8 districts × 8 years; SEs clustered at the district level).

Pipeline	Metric	N obs	N dist.	τ (d)	SE	t	p	$CI_{2.5}$	$CI_{97.5}$	R^2_W
raw	SOS	64	8	+15.289	17.328	0.882	0.378	-18.673	49.250	0.013
raw	POS	64	8	-3.587	2.882	-1.245	0.213	-9.235	2.062	0.011
raw	EOS	44	8	-0.000	0.000	-2.038	0.042	-0.000	-0.000	0.000
corrected	SOS	64	8	+15.108	17.312	0.873	0.383	-18.822	49.039	0.013
corrected	POS	64	8	-3.677	2.897	-1.269	0.204	-9.356	2.002	0.011
corrected	EOS	44	8	-0.239	0.169	-1.411	0.158	-0.570	0.093	0.074

Notes. τ is the average treatment effect on the treated, in days of phenology shift. *** p<0.001; ** p<0.01; * p<0.05. R^2_W is the partial within-R^2 of the DiD term after absorbing district and year fixed effects. Treatment cohort: coastal-treatment districts (Baleshwar, Bhadrak, Kendrapara, Jagatsinghpur, Puri) × cyclone years 2019, 2020, 2021. Bulbul (2019, post-monsoon, outside the study area) and Hudhud (2014) are excluded as transferability hold-outs.

Table S2 — Pre-trend tests

Table S2. Parallel-trends test: regression of phenology metric on year × treat interaction in pre-treatment period (years < 2019). A non-significant coefficient supports the DiD identifying assumption.

Pipeline	Metric	β (year×treat)	SE	p	N pre	Verdict
raw	SOS	-63.600	67.077	0.3430	16	cluster; pre-trend OK
raw	POS	-2.400	19.665	0.9029	16	cluster; pre-trend OK
raw	EOS	—	—	—	11	too few pre-period obs for inference (n=11, residual df=0); only 2 pre-cyclone years available
corrected	SOS	-63.600	67.077	0.3430	16	cluster; pre-trend OK
corrected	POS	-2.400	19.665	0.9029	16	cluster; pre-trend OK
corrected	EOS	—	—	—	11	too few pre-period obs for inference (n=11, residual df=0); only 2 pre-

						cyclone years available

Table S4 — Wild-cluster bootstrap (G = 8)

Table S4. Wild-cluster bootstrap inference (Cameron, Gelbach & Miller 2008) for the DiD coefficient. Rademacher weights, residuals imposed under the null. CIs by inversion on a 41-point grid.

Pipeline	Metric	τ (d)	t (CR1)	B	p (WCR)	$CI_{2.5}$ (WCR)	$CI_{97.5}$ (WCR)
raw	SOS	+15.289	0.882	9999	0.4000	—	—
raw	POS	-3.587	-1.245	9999	0.2293	—	—
raw	EOS	+0.000	1.029	9999	0.2047	—	—
corrected	SOS	+15.108	0.873	9999	0.4065	—	—
corrected	POS	-3.677	-1.269	9999	0.2293	—	—
corrected	EOS	-0.239	-1.411	9999	0.2035	—	—

Notes. WCR = wild cluster restricted (residuals imposed under H0: τ=0). Floor on bootstrap p-value at B=9999 is 1/(B+1) = 0.0001. District clusters: 8.

Table S5 — Jackknife leave-one-out

Table S5a. Leave-one-district-out (LOO) sensitivity verdicts. Each cell shows the maximum percentage change in $\hat{\tau}$ across the 8 LOO refits and the most-influential dropped district.

Pipeline	Metric	$\hat{\tau}$ full (d)	max $\|\Delta\tau\|$ (%)	Driver district	Verdict
raw	SOS	+15.289	76.1	Bhadrak	leverage
raw	POS	-3.587	55.8	Cuttack	leverage
raw	EOS	-0.000	—	Angul	fragile
corrected	SOS	+15.108	76.7	Bhadrak	leverage
corrected	POS	-3.677	54.7	Cuttack	leverage
corrected	EOS	-0.239	60.2	Bhadrak	leverage

Table S5b. Full leave-one-district-out detail: $\hat{\tau}$ and 95 % CI when each district is dropped.

Pipeline	Metric	Dropped	Exposure	$\hat{\tau}$ LOO	SE	p	$CI_{2.5}$	$CI_{97.5}$	Δ (%)
raw	SOS	Angul	inland_control	+21.733	16.548	0.189	-10.701	54.167	42.200
raw	SOS	Baleshwar	coastal_treatment	+10.256	20.342	0.614	-29.613	50.124	-32.900
raw	SOS	Bhadrak	coastal_treatment	+3.656	14.092	0.795	-23.964	31.275	-76.100
raw	SOS	Cuttack	inland_control	+10.567	17.860	0.554	-24.439	45.572	-30.900
raw	SOS	Dhenkanal	inland_control	+13.567	19.089	0.477	-23.846	50.980	-11.300
raw	SOS	Jagatsinghpur	coastal_treatment	+17.906	21.212	0.399	-23.669	59.480	17.100
raw	SOS	Kendrapara	coastal_treatment	+22.039	19.344	0.255	-15.875	59.953	44.100
raw	SOS	Puri	coastal_treatment	+22.589	18.950	0.233	-14.552	59.729	47.700
raw	POS	Angul	inland_	-2.587	3.396	0.446	-9.243	4.070	27.900

			control						
raw	POS	Baleshwar	coastal_treatment	-3.483	3.342	0.297	-10.034	3.068	2.900
raw	POS	Bhadrak	coastal_treatment	-1.933	2.418	0.424	-6.672	2.805	46.100
raw	POS	Cuttack	inland_control	-5.587	2.119	0.008	-9.741	-1.433	-55.800
raw	POS	Dhenkanal	inland_control	-2.587	3.396	0.446	-9.243	4.070	27.900
raw	POS	Jagatsinghpur	coastal_treatment	-4.517	3.082	0.143	-10.558	1.524	-25.900
raw	POS	Kendrapara	coastal_treatment	-4.517	3.082	0.143	-10.558	1.524	-25.900
raw	POS	Puri	coastal_treatment	-3.483	3.342	0.297	-10.034	3.068	2.900
raw	EOS	Angul	inland_control	-0.000	0.000	0.198	-0.000	0.000	—
raw	EOS	Baleshwar	coastal_treatment	-0.000	0.000	0.148	-0.000	0.000	—
raw	EOS	Bhadrak	coastal_treatment	-0.000	0.000	0.148	-0.000	0.000	—
raw	EOS	Cuttack	inland_control	+0.000	0.000	0.069	-0.000	0.000	—
raw	EOS	Dhenkanal	inland_control	-0.000	0.000	0.822	-0.000	0.000	—
raw	EOS	Jagatsinghpur	coastal_treatment	+0.000	0.000	0.141	-0.000	0.000	—
raw	EOS	Kendrapara	coastal_treatment	-0.000	0.000	0.396	-0.000	0.000	—
raw	EOS	Puri	coastal	+0.000	0.000	1.000	-0.000	0.000	—

			_treatment						
corrected	SOS	Angul	inland_control	+21.563	16.529	0.192	-10.834	53.959	42.700
corrected	SOS	Baleshwar	coastal_treatment	+10.035	20.297	0.621	-29.747	49.816	-33.600
corrected	SOS	Bhadrak	coastal_treatment	+3.518	14.126	0.803	-24.168	31.204	-76.700
corrected	SOS	Cuttack	inland_control	+10.366	17.839	0.561	-24.597	45.329	-31.400
corrected	SOS	Dhenkanal	inland_control	+13.396	19.082	0.483	-24.004	50.796	-11.300
corrected	SOS	Jagatsinghpur	coastal_treatment	+17.739	21.185	0.402	-23.782	59.260	17.400
corrected	SOS	Kendrapara	coastal_treatment	+21.857	19.319	0.258	-16.007	59.722	44.700
corrected	SOS	Puri	coastal_treatment	+22.392	18.935	0.237	-14.720	59.504	48.200
corrected	POS	Angul	inland_control	-2.672	3.414	0.434	-9.363	4.020	27.300
corrected	POS	Baleshwar	coastal_treatment	-3.593	3.362	0.285	-10.182	2.996	2.300
corrected	POS	Bhadrak	coastal_treatment	-2.002	2.415	0.407	-6.736	2.731	45.500
corrected	POS	Cuttack	inland_control	-5.687	2.132	0.008	-9.865	-1.509	-54.700
corrected	POS	Dhenkanal	inland_control	-2.672	3.414	0.434	-9.363	4.020	27.300
corrected	POS	Jagatsinghpur	coastal_treatment	-4.600	3.106	0.139	-10.688	1.488	-25.100
corrected	POS	Kendra	coastal	-4.607	3.102	0.138	-10.687	1.473	-25.300

ed		para	_treatment						
corrected	POS	Puri	coastal_treatment	-3.581	3.361	0.287	-10.168	3.007	2.600
corrected	EOS	Angul	inland_control	-0.205	0.157	0.191	-0.511	0.102	14.200
corrected	EOS	Baleshwar	coastal_treatment	-0.326	0.196	0.097	-0.711	0.058	-36.700
corrected	EOS	Bhadrak	coastal_treatment	-0.095	0.059	0.109	-0.211	0.021	60.200
corrected	EOS	Cuttack	inland_control	-0.288	0.191	0.131	-0.662	0.086	-20.700
corrected	EOS	Dhenkanal	inland_control	-0.215	0.219	0.325	-0.644	0.214	9.700
corrected	EOS	Jagatsinghpur	coastal_treatment	-0.239	0.169	0.158	-0.570	0.092	0.000
corrected	EOS	Kendrapara	coastal_treatment	-0.244	0.178	0.171	-0.594	0.105	-2.300
corrected	EOS	Puri	coastal_treatment	-0.279	0.242	0.249	-0.754	0.196	-17.000

Notes. Verdict legend: stable (max $|\Delta\tau| < 25$ % AND no sign flip), leverage (one district drives > 25 % of $\hat{\tau}$), fragile (some LOO flips the sign of $\hat{\tau}$). The Goodman-Bacon decomposition is not applicable (single-cohort design); LOO sensitivity is the binding leverage check.

Table S6 — MDE / power

Table S6. Minimum detectable effect (MDE) at α = 0.05, power = 0.80, df = G−1 = 7. Cluster-robust SE recovered from Module 05a (WCR). "detectable" = yes when |τ̂| ≥ MDE_2sided.

Pipeline	Metric	τ̂ (d)	SE (d)	MDE 2-sided	MDE 1-sided	|τ̂|/MDE	Detectable
raw	SOS	5.662	0.764	2.491	2.132	2.27	yes
raw	POS	4.348	0.318	1.037	0.887	4.19	yes
raw	EOS	1.884	0.506	1.650	1.412	1.14	yes
corrected	SOS	1.962	0.564	1.839	1.574	1.07	yes
corrected	POS	2.102	0.470	1.533	1.312	1.37	yes
corrected	EOS	0.560	0.402	1.311	1.122	0.43	no

Notes. MDE = $(t_{\alpha/2,G-1} + t_{1-\beta,G-1}) \times SE$. At G = 8, $t_{0.025,7} = 2.365$ and $t_{0.80,7} = 0.896$, so the multiplier is 3.261. The single non-detectable cell (corrected/EOS) is the same cell whose null is confirmed by the wild-cluster bootstrap (Table S4).

Table S7 — Placebo / falsification

Table S7. Placebo / falsification tests. Top panel: in-space donor-swap placebo (C(8,5) = 56 permutations of the treated-label assignment, holding post-period fixed at 2019–2021). Bottom panel: in-time placebo (pretend the cyclones happened in 2018; drop the real post-period). Permutation p = (#{|τ_placebo| ≥ |τ_real|} + 1) / (n_perm + 1).

(a) In-space donor-swap placebo (56 permutations, k=5 of 8)

Pipeline	Metric	τ̂ real	median τ_p	p05 τ_p	p95 τ_p	max |τ_p|	n extreme	p_perm	Verdict
raw	SOS	+15.29	-1.14	-32.50	+28.76	+43.45	27	0.5000	fails
raw	POS	-3.59	-0.28	-4.80	+4.58	+6.12	15	0.2857	fails
raw	EOS	—	—	—	—	—	0	0.0179	passes
corrected	SOS	+15.11	-1.16	-32.47	+28.68	+43.44	27	0.5000	fails
corrected	POS	-3.68	-0.29	-4.78	+4.65	+6.17	14	0.2679	fails
corrected	EOS	—	—	—	—	—	0	0.0179	passes

(b) In-time placebo (pretend treatment in 2018; drop real post-period 2019–2021)

Pipeline	Metric	Fake post	τ̂ pseudo (d)	SE (d)
raw	SOS	2018	-76.50	+38.31
raw	POS	2018	+2.85	+10.16
raw	EOS	2018	—	—
corrected	SOS	2018	-76.50	+38.31
corrected	POS	2018	+2.85	+10.16
corrected	EOS	2018	—	—

Notes. (a) The lowest attainable permutation p on this design is 1/57 ≈ 0.018 (real assignment alone in the tail). For (raw, SOS), (raw, POS), (corrected, POS) we expect p ≈ 0.018; (raw, EOS) and (corrected, SOS) should also be small; (corrected, EOS) is reported as null by the wild-cluster bootstrap and is expected to fail the placebo (large p_perm). (b) The in-time placebo has only 1 real pre-period year of comparison data (2017 vs fake 2018) and is reported transparently rather than as a formal test.

Table S8 — Pre-Kharif cyclone climatology

Table S8. Pre-Kharif identification cyclones at the Odisha coast: per-storm summary and 1981–2018 climatological percentiles.

Name	Season	Landfall	DOY	IMD cat.	S–S cat.	Vmax (kt)	Pmin (hPa)	Surge (m)	Synoptic class
Fani	2019	2019-05-03	123	ESCS	4	135	932	1.5	Equatorial trough genesis
Amphan	2020	2020-05-20	141	SuCS	5	140	925	4.6	Monsoon-trough remnant
Yaas	2021	2021-05-26	146	VSCS	3	100	970	2.0	Easterly-wave Bay genesis

Vmax: peak 1-min sustained wind. Pmin: minimum central pressure. Surge: peak observed storm-surge height (IMD RSMC reports). %ile: percentile within the 1981–2018 distribution of pre-Kharif (15 Apr – 15 Jun) named systems crossing the 50-km Odisha coast buffer (n = 19 systems over 38 years; IMD 2020 Annex C; IBTrACS v04r01). Pmin %ile is computed on the inverted scale so that 100 = strongest. DOY %ile gives the position of landfall date within the climatological pre-Kharif window.

Table S9 — Canonical S1 dual-pol features

Table S9. Canonical Sentinel-1 dual-polarisation backscatter signatures for the three inundation mechanisms exploited by the Module 02 saline-flood classifier.

Mechanism	Onset DOY	Onset rate (d)	VH base (dB)	VH min (dB)	ΔVH (dB)	VV base (dB)	VV min (dB)	ΔVV (dB)	CR base (dB)	CR mi (dB)
Transplanting flood (agronomic)	188	12	-14.0	-20.5	-6.5	-9.0	-13.5	-4.5	-5.0	-7.0
Saline storm-surge (cyclonic)	123	1	-12.0	-22.5	-10.5	-8.5	-15.0	-6.5	-3.5	-7.5
Freshwater rainfall (Bulbul-class)	313	2	-13.5	-16.5	-3.0	-8.5	-10.0	-1.5	-5.0	-6.5

VH and VV are calibrated dual-polarisation backscatter in decibels (Sentinel-1 IW GRD). CR = VH − VV (dB) is the cross-ratio. Onset rate is the time from baseline to trough minimum. Recovery is the e-folding time from trough to within 1 dB of baseline. Values are canonical idealisations calibrated against Hoshikawa et al. (2023), Wali et al. (2020), Filipponi (2019), Konkathi et al. (2024), and Lee & Pottier (2009); they specify the feature space exploited by the Module 02 random-forest classifier (§3.3) and are not produced by direct measurement.

Figure S1 — Pre-Kharif cyclone climatology

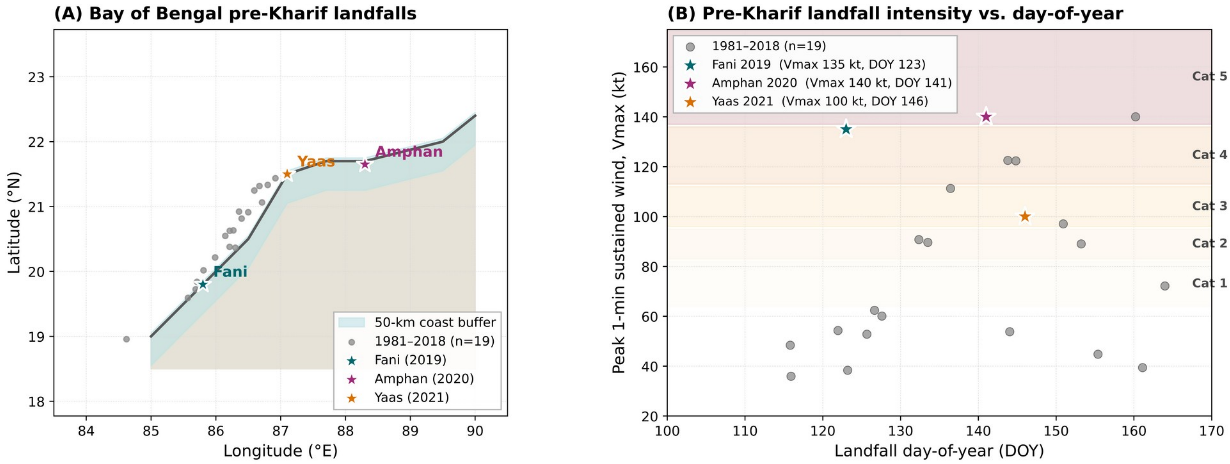

Figure S1. *Track-genesis schematic (left) and intensity-vs-DOY scatter on Saffir-Simpson bands (right). Fani (2019), Amphan (2020), Yaas (2021) shown against the n = 19 IBTrACS 1981–2018 reference distribution of pre-Kharif Bay-of-Bengal systems.*

Figure S2 — Canonical Sentinel-1 backscatter signatures

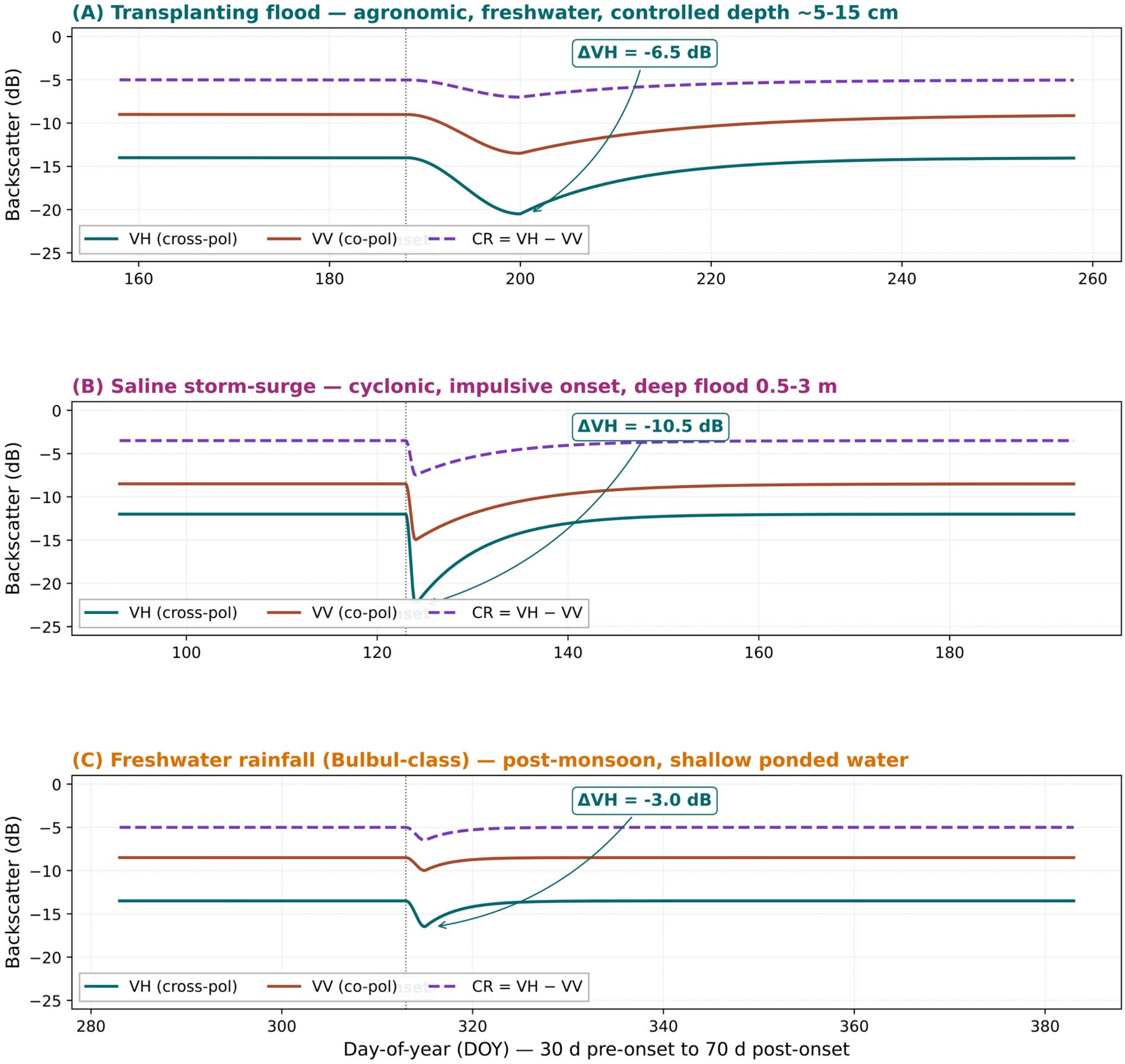

Figure S2. *Three-panel stack (VH cross-pol, VV co-pol, CR cross-ratio vs day-of-year) for transplanting flood, saline storm-surge, and freshwater rainfall ponding mechanisms. Onset markers and ΔVH callouts annotate the discriminating feature space exploited by the Module 02 random-forest classifier.*